Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2026-27

Class:	TE-AI&DS	Semester:	V
Course Code:	2015118	Course Name:	Artificial Intelligence & Soft Computing Lab

Name of Student:	Rushikesh Subhash Khemnar
Roll No. :	81
Experiment No.:	9
Title of the Experiment:	Implement a Fuzzy Logic Controller for Real-World Applications
Date of Performance:	28/09/26
Date of Submission:	05/10/26

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty : Signature :

Date :

Aim: To implement a simple Fuzzy Logic Controller for real-world control problems.

Objectives:

After completing this experiment, the student will be able to:

1. To understand the fundamentals of Fuzzy Logic and Fuzzy Inference Systems.
2. To study the components of a Fuzzy Logic Controller (FLC).
3. To implement a simple Fuzzy Logic Controller for a real-world application.
4. To design fuzzy membership functions and rule base.
5. To analyze the performance of the Fuzzy Logic Controller.

Tools Required:

- Hardware:
  - Desktop/Laptop
- Software:
  - Python 3.x with Scikit-Fuzzy Library / MATLAB Fuzzy Logic Toolbox
  - Visual Studio Code / Jupyter Notebook
  - Windows/Linux Operating System

Theory:

Fuzzy Logic is an Artificial Intelligence technique that deals with reasoning under uncertainty by allowing variables to have degrees of truth instead of binary values. Unlike classical logic, where values are restricted to either 0 or 1, fuzzy logic represents information using linguistic variables such as Low, Medium, and High. This enables intelligent systems to make decisions in situations where precise mathematical models are difficult to develop.

A Fuzzy Logic Controller (FLC) consists of four major components: Fuzzification, Knowledge Base, Inference Engine, and Defuzzification. Fuzzification converts crisp input values into fuzzy values, the inference engine applies predefined IF–THEN rules to make decisions and defuzzification converts the fuzzy output into a crisp value. Fuzzy Logic Controllers are widely used in industrial automation, air conditioners, washing machines, automotive systems and process control applications.

Procedure:

1. Select a suitable real-world control problem.
2. Define the input and output variables for the controller.
3. Design appropriate fuzzy membership functions. 4. Create the fuzzy rule base using IF–THEN rules.
5. Implement the Fuzzy Logic Controller using the selected software tool.
6. Execute the controller for different input values.
7. Observe and analyze the generated output.

Description of the Experiment:

This experiment demonstrates the implementation of a simple Fuzzy Logic Controller for solving a real-world control problem. Suitable input variables and membership functions are defined and a set of fuzzy IF–THEN rules is created to model the control system.

The experiment illustrates how fuzzy reasoning is used to process uncertain or imprecise inputs and generate appropriate control outputs without requiring an exact mathematical model of the system.

Detailed Description:

A Fuzzy Logic Controller is developed by defining suitable input and output variables along with their corresponding fuzzy membership functions. The relationship between the input and output variables is represented through a set of fuzzy IF–THEN rules, which form the knowledge base of the controller. The inference engine evaluates these rules based on the given inputs and produces fuzzy outputs, which are then converted into crisp values through the defuzzification process.

The implemented controller is tested using different input conditions to observe its decisionmaking capability. The generated outputs are analyzed to evaluate the effectiveness of fuzzy reasoning in handling uncertainty and controlling real-world systems without relying on precise mathematical models.

Code:- import numpy
as np import skfuzzy
as fuzz
from skfuzzy import control as ctrl

temperature = ctrl.Antecedent(np.arange(0, 41, 1), 'temperature')

```
fan_speed = ctrl.Consequent(np.arange(0, 101, 1), 'fan_speed')

temperature['low'] = fuzz.trimf(temperature.universe, [0, 0, 20])
temperature['medium'] = fuzz.trimf(temperature.universe, [10, 20, 30])
temperature['high'] = fuzz.trimf(temperature.universe, [20, 40, 40])

fan_speed['low'] = fuzz.trimf(fan_speed.universe, [0, 0, 50])
fan_speed['medium'] = fuzz.trimf(fan_speed.universe, [25, 50, 75])
fan_speed['high'] = fuzz.trimf(fan_speed.universe, [50, 100, 100])

rule1 = ctrl.Rule(temperature['low'], fan_speed['low']) rule2
= ctrl.Rule(temperature['medium'], fan_speed['medium'])
rule3 = ctrl.Rule(temperature['high'], fan_speed['high'])

fan_control = ctrl.ControlSystem([rule1, rule2, rule3])
fan_simulation = ctrl.ControlSystemSimulation(fan_control)
test_temperatures = [10, 20, 25, 30, 35, 40]

print("FUZZY LOGIC FAN SPEED CONTROLLER")
print("---------------------------------")

for temp in test_temperatures:
fan_simulation.input['temperature'] = temp
fan_simulation.compute()

    speed = fan_simulation.output['fan_speed']

    print(f"Temperature: {temp} °C  ->  Fan Speed: {speed:.2f}%")

temp = float(input("\nEnter temperature (0-40 °C): "))

fan_simulation.input['temperature'] = temp
fan_simulation.compute()

print(f'Fan Speed: {fan_simulation.output['fan_speed']:.2f}%")
```

Output:-

```
FUZZY LOGIC FAN SPEED CONTROLLER
--------------------------------
Temperature: 10 °C  ->  Fan Speed: 19.44%
Temperature: 20 °C  ->  Fan Speed: 50.00%
Temperature: 25 °C  ->  Fan Speed: 59.79%
Temperature: 30 °C  ->  Fan Speed: 80.56%
Temperature: 35 °C  ->  Fan Speed: 82.50%
Temperature: 40 °C  ->  Fan Speed: 83.33%

Enter temperature (0-40 °C): 26
Fan Speed: 62.45%
```

Conclusion:

A simple Fuzzy Logic Controller was successfully implemented for a real-world control problem using fuzzy membership functions and rule-based inference. The experiment demonstrated the effectiveness of fuzzy logic in handling uncertainty and generating intelligent control decisions.

Questions to be Answered by Students:

1.Which real-world control problem was selected for implementing the Fuzzy Logic Controller?

Real-world control problem: Fan speed control based on room temperature.

2.Which input and output variables were considered in the fuzzy system?

Input and output variables:

- Input: Temperature (°C)
- Output: Fan Speed (%)

3.How many fuzzy IF–THEN rules were defined for the controller?

Number of fuzzy IF–THEN rules: 3 rules

- If temperature is Low → Fan speed is Low
- If temperature is Medium → Fan speed is Medium
- If temperature is High → Fan speed is High

4.State two observations obtained from the implementation of the Fuzzy Logic Controller

Two observations:

- As the temperature increases, the fan speed also increases.

- The Fuzzy Logic Controller provides a smooth fan-speed response for different temperature conditions.

.